

PROGRAM 9:Organize the kubernets pods using the Labels and Namespaces

Step1: create a file as namespaces.[yaml](#)

```
GNU nano 7.2
apiVersion: v1
kind: Namespace
metadata:
  name: dev
---
apiVersion: v1
kind: Namespace
metadata:
  name: prod
```

Step2: create a file as pods.yaml

GNU nano 7.2

```
apiVersion: v1
kind: Pod
metadata:
  name: frontend-pod
  namespace: dev
  labels:
    app: frontend
    env: dev
spec:
  containers:
  - name: frontend
    image: nginx
    ports:
    - containerPort: 80
```

```
apiVersion: v1
kind: Pod
metadata:
  name: backend-pod
  namespace: dev
  labels:
    app: backend
    env: dev
spec:
  containers:
  - name: backend
    image: nginx
    ports:
    - containerPort: 80
```

```
apiVersion: v1
kind: Pod
metadata:
  name: frontend-pod
  namespace: prod
  labels:
    app: frontend
    env: prod
```

```
spec:
  containers:
  - name: frontend
    image: nginx
    ports:
```

Step3: apply the namespaces and pods yaml file

>kubectl apply -f namespaces.yaml

>kubectl apply -f pods.yaml

Step4: >kubectl get pods -n dev

>kubectl get pods -n prod

Step5: check pod status

>kubectl get pod

Step6: filter pods status

>kubectl get pods -n dev -l app=fronted

>kubectl get pods -n prod -l app=backend

```
configuring bridge CNI (Container Networking Interface) ...
Verifying Kubernetes components...
  ■ Using image gcr.io/k8s-minikube/storage-provisioner:v5
  ■ Enabled addons: storage-provisioner, default-storageclass
Done! kubectl is now configured to use "minikube" cluster and "default" namespace by default
1RV24MC104_SINDHURA@sindhra-IdeaPad-Slim-5-16IAH8:~$ kubectl get nodes
NAME                STATUS    ROLES    AGE   VERSION
minikube            Ready     control-plane   11s   v1.34.0
1RV24MC104_SINDHURA@sindhra-IdeaPad-Slim-5-16IAH8:~$ kubectl config use-context minikube
Switched to context "minikube".
1RV24MC104_SINDHURA@sindhra-IdeaPad-Slim-5-16IAH8:~$ mkdir prog9
1RV24MC104_SINDHURA@sindhra-IdeaPad-Slim-5-16IAH8:~$ cd prog9
1RV24MC104_SINDHURA@sindhra-IdeaPad-Slim-5-16IAH8:~/prog9$ nano namespaces.yaml
1RV24MC104_SINDHURA@sindhra-IdeaPad-Slim-5-16IAH8:~/prog9$ nano pods.yaml
1RV24MC104_SINDHURA@sindhra-IdeaPad-Slim-5-16IAH8:~/prog9$ kubectl apply -f namespaces.yaml
namespace/dev created
namespace/prod created
1RV24MC104_SINDHURA@sindhra-IdeaPad-Slim-5-16IAH8:~/prog9$ kubectl apply -f pods.yaml
error: error parsing pods.yaml: error converting YAML to JSON: yaml: line 56: did not find expected key
1RV24MC104_SINDHURA@sindhra-IdeaPad-Slim-5-16IAH8:~/prog9$ nano pods.yaml
1RV24MC104_SINDHURA@sindhra-IdeaPad-Slim-5-16IAH8:~/prog9$ kubectl apply -f pods.yaml
pod/backend-pod created
1RV24MC104_SINDHURA@sindhra-IdeaPad-Slim-5-16IAH8:~/prog9$ kubectl get pods -n dev
No resources found in dev namespace.
1RV24MC104_SINDHURA@sindhra-IdeaPad-Slim-5-16IAH8:~/prog9$ kubectl get namespaces
NAME                STATUS    AGE
default             Active    24m
```

```

1RV24MC104_SINDHURA@sindhra-IdeaPad-Slim-5-16IAH8:~/prog9$ kubectl apply -f pods.yaml
pod/frontend-pod created
pod/backend-pod created
pod/frontend-pod created
pod/backend-pod created
1RV24MC104_SINDHURA@sindhra-IdeaPad-Slim-5-16IAH8:~/prog9$ kubectl get pods -n dev
NAME          READY   STATUS             RESTARTS   AGE
backend-pod    0/1     ContainerCreating   0           11s
frontend-pod   1/1     Running             0           11s
1RV24MC104_SINDHURA@sindhra-IdeaPad-Slim-5-16IAH8:~/prog9$ kubectl get pods -n dev
NAME          READY   STATUS    RESTARTS   AGE
backend-pod    1/1     Running   0           51s
frontend-pod   1/1     Running   0           51s
1RV24MC104_SINDHURA@sindhra-IdeaPad-Slim-5-16IAH8:~/prog9$ kubectl get pods -n prod
NAME          READY   STATUS    RESTARTS   AGE
backend-pod    1/1     Running   0           75s
frontend-pod   1/1     Running   0           75s
1RV24MC104_SINDHURA@sindhra-IdeaPad-Slim-5-16IAH8:~/prog9$ kubectl get pods -n dev -l app=frontend
NAME          READY   STATUS    RESTARTS   AGE
frontend-pod   1/1     Running   0           91s
1RV24MC104_SINDHURA@sindhra-IdeaPad-Slim-5-16IAH8:~/prog9$ kubectl get pods -n prod -l app=backend
NAME          READY   STATUS    RESTARTS   AGE
backend-pod    1/1     Running   0          111s
1RV24MC104_SINDHURA@sindhra-IdeaPad-Slim-5-16IAH8:~/prog9$
1RV24MC104_SINDHURA@sindhra-IdeaPad-Slim-5-16IAH8:~/prog9$
1RV24MC104_SINDHURA@sindhra-IdeaPad-Slim-5-16IAH8:~/prog9$

```